Feasibility Review: Synthetic Prostate MRI with VQ-VAE-2 (HipMRI)

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Open dilemma: can a generative model produce prostate MRI slices that are realistic and diverse **without memorising** the training patients?

1. User need, scope and acceptance criteria

Intended user: a research group holding the HipMRI study that wants a synthetic cohort it can share or use for augmentation without exposing real patients. **Prototype purpose:** test whether a VQ-VAE-2 with learned priors meets that need, and how it compares with GAN baselines. **Scope:** unconditional 2D slices (256×128), no segmentation labels.

1. **Fidelity:** mean SSIM > 0.6 between test slices and their reconstructions.
2. **Quality:** FID / KID against unseen patients, reported next to a real-data floor and the DCGAN / WGAN-GP baselines.
3. **No memorisation:** ≤ 5 % of samples are *near-copies* (closer to a training slice than any real unseen slice is).
4. **Diversity:** pairwise SSIM between samples no higher than between real slices (no mode collapse).
5. **Resource constraint:** ≤ 2 h training per model and ≤ 8 GB peak VRAM on one A100.

2. Model choice and course concepts

- **Dataset configuration:** HipMRI 2D slices, 12,660 slices from 38 patients, split **31 / 4 / 3 by patient** (no leakage). Settings are chosen on validation; test is used once.
- **Model (Hard band):** VQ-VAE-2 encodes each slice into two grids of discrete codes: a 16×32 top grid for layout and a 32×64 bottom grid for detail, each with a 512-entry codebook. Two gated PixelCNN priors learn the code grids; sampling the priors and decoding gives new slices.
- **Course concepts:** convolutional autoencoders, vector quantisation, autoregressive models (masked convolutions), GANs, and regularisation (dropout, weight decay, early stopping, augmentation).
- **Literature:** VQ-VAE (van den Oord et al., 2017); VQ-VAE-2 (Razavi et al., 2019); gated PixelCNN (van den Oord et al., 2016); DCGAN (Radford et al., 2016); WGAN-GP (Gulrajani et al., 2017); FID (Heusel et al., 2017). The two-level design targets the known weakness of single-level priors: correct texture but incoherent global anatomy.

3. Preliminary feasibility evidence

- **Data audit:** all 38 patients load; zero patient overlap between splits (checked in code).
- **Reconstruction:** test SSIM **0.92** (every slice > 0.6) with 507 / 512 codes in use. **Criterion 1 met.**
- **Smoke test of generation:** the two-level model is about 45 FID better than a single-level VQ-VAE, with visibly coherent anatomy.
- **Memorisation:** about 1 % near-copies for VQ-VAE-2, against 9.4 % for WGAN-GP.
- **Measured resources:** every model trains in under 2 h at ≤ 4.7 GB VRAM. **Criterion 5 met.**

4. Risks, budget and fallback

Risk	Mitigation
Prior overfits 31 patients	Dropout, weight decay, left-right flip augmentation, early stopping, best checkpoint
Sampling is slow (about 1.5 s per slice)	Accept for offline cohort generation; a faster prior is future work
A GAN baseline scores better FID	Report honestly; recommend a model per use case
Codebook collapse	Dead-code restarts; monitor codes used

Computing budget: about 15 A100 GPU-hours on Rangpur in total; only best checkpoints are kept.
Next planned experiment: one-off test-set benchmark of the selected configurations (3 seeds × 256 samples).
Fallback: WGAN-GP or DCGAN with the same evaluation (Normal band).